STA 4001 Stochastic Process, Fall 2021
Midterm (October 23, 2-4 pm)

1. (20 points) 0.5 percent of the population has a particular disease. A test is developed for the disease. The test gives a false positive 3% of the time and a false negative 2% of the time.

(a) (10 points) What is the probability that a random person tests positive?

(b) (10 points) Joe just got the bad news that the test came out positive; what is the probability that Joe has the disease?

Solution:

(a) Let D be the event that a random person has this disease, and let P represent the event that the test for this person turns out positive. From the description of this question, we know that $\mathbb{P}(D) = 0.005$, $\mathbb{P}(P|D^c) = 0.03$ and $\mathbb{P}(P^c|D) = 0.02$.

$$\begin{aligned}\mathbb{P}(P) &= \mathbb{P}(PD) + \mathbb{P}(PD^c) \\ &= \mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^c)\mathbb{P}(D^c) \\ &= (1 - 0.02) \times 0.005 + 0.03 \times (1 - 0.005) \\ &= 0.0347\end{aligned}$$

(b) We need to calculate $\mathbb{P}(D|P)$. By definition of conditional probability, we have

$$\begin{aligned}\mathbb{P}(D|P) &= \frac{\mathbb{P}(D)\mathbb{P}(P|D)}{\mathbb{P}(P)} \\ &= \frac{0.98 \times 0.005}{0.0347} \\ &= 0.1412.\end{aligned}$$

2. (20 points) Consider a Markov chain $X_n \in \{0, 1, 2, 3, 4, 5, 6\}$ with transition matrix

$$P = \begin{Vmatrix} 1/2 & 0 & 1/8 & 1/4 & 1/8 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 \end{Vmatrix}.$$

1. (10 points) Specify the classes of this Markov chain, and give the reason.
2. (10 points) Determine whether these classes are transient or recurrent, and give the reason.

Solution:

1. Figure 1 is the transition graph of X_n. From the transition graph, we can see that
   (a) State 1 is not accessible from any other states, so $\{1\}$ is a class.
   (b) State 0,2,3 communicate with each other. They are not accessible from state 4,5,6, and they cannot reach state 1. So $\{0, 2, 3\}$ is a class.
   (c) State 4,5,6 communicate with each other. They cannot reach any other states. So $\{4, 5, 6\}$ is a class.

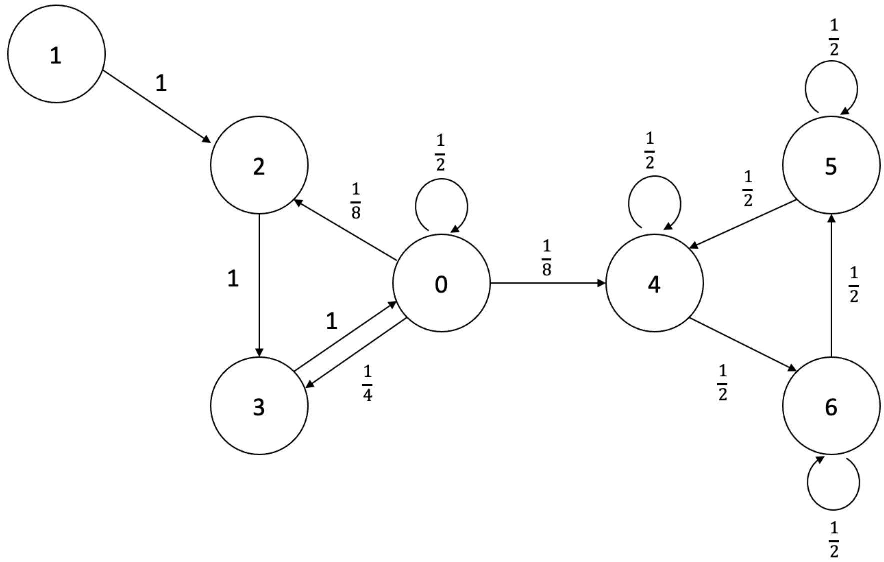


Figure 1: Transition graph of X_n in Problem 2.

In summary, there are 3 classes: $\{1\},\ \{0,2,3\},\ \{4,5,6\}$.

2. The class $\{1\}$ is transient because $\mathbb{P}(X_n = 1|X_0 = 1) = 0$ for all n.
   The class $\{0,2,3\}$ is also transient. Suppose $X_0 = 0$ and $X_1 = 4$, then this Markov chain never goes back to state 0. Thus,

$$\begin{aligned} 1 - f_0 &= \mathbb{P}(X_n \neq 0 \text{ for all } n|X_0 = 0) \\ &\geq \mathbb{P}(X_1 = 4|X_0 = 0) = \frac{1}{8}. \end{aligned}$$

   So $f_0 \leq \frac{7}{8} < 1$. As a result, class $\{0,2,3\}$ is transient.
   Class $\{4,5,6\}$ is recurrent. Since this Markov chain only has 7 states, not all states can be transient. Because class $\{1\}$ and class $\{0,2,3\}$ are transient, the only one class left has to be recurrent.

3. (20 points) Suppose to make a product involves two steps. After Step 1, with probability 10%, the product is damaged and will be discarded, with probability 20%, we need redo Step 1 for the product, and with probability 70%, the product can be sent to Step 2. After Step 2, with probability 5%, the product is damaged and will be discarded, with probability 5%, we need redo Step 1 for the product, with probability 10%, we need redo Step 2, and with probability 80%, the product is made.

(a) (10 points) Construct a Markov chain with two absorbing states to model the above production procedure.

(b) (10 points) Find out the probability that a product is damaged in the above procedure.

Solution:

(a) We can construct a Markov chain $\{X_n\}$ with 4 states, and $X_n \in \{0,1,2,3\}$, which denotes that a product is damaged and will be discarded, is at Step 1, is at Step 2, and is made, respectively. Then state 0 and 3 are two absorbing states, and the corresponding transition matrix is:

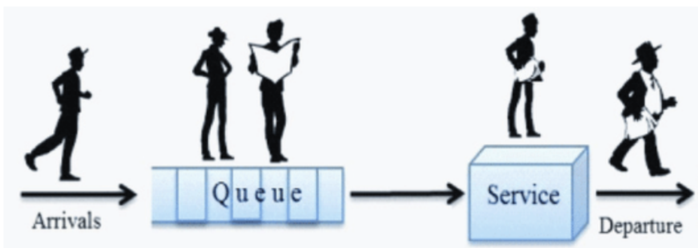
$$P = \begin{Vmatrix} 1 & 0 & 0 & 0 \\ 0.1 & 0.2 & 0.7 & 0 \\ 0.05 & 0.05 & 0.1 & 0.8 \\ 0 & 0 & 0 & 1 \end{Vmatrix}.$$

(b) Let P_i be the probability that starting from state i, the DTMC is absorbed into state 0. Then our objective is to find out P_1, and we have:

$$\begin{aligned}
&P_0 = 1\\
&P_1 = 0.1P_0 + 0.2P_1 + 0.7P_2\\
&P_2 = 0.05P_0 + 0.05P_1 + 0.1P_2 + 0.8P_3\\
&P_3 = 0
\end{aligned}$$

Solving this system of linear equations, we can get $P_1 = \frac{25}{137}$.

4. (40 points) In this question, we shall analyze the long-run behavior of a single-server queue in discrete time. Consider a service station with a single server. Suppose in each time step, the number of new arrivals follows a Bernoulli distribution with parameter $\lambda \in (0, 1)$ and arrivals in different time steps are independent. The service time of each customer follows a geometric distribution with parameter $\mu \in (0, 1)$ and services times of different customers are independent. Also, we assume that arrivals and service times are independent. The server takes customers in the first-come-first-serve manner and can only serve one customer at a time. Customers will leave the station immediately after the service is finished. Let's denote by X_n the number of customers in the service station (including the one in service) at time n.



(a) (10 points) Show that X_n is a DTMC and find its transition probabilities.

(b) (10 points) Show that X_n is time reversible when $\lambda = 0.6$ and $\mu = 0.9$. Find out its stationary distribution.

(c) (8 points) Show that X_n does not have a stationary distribution when $\lambda = 0.9$ and $\mu = 0.6$.

(d) (2 points) What will happen to X_n in (c) when $n \to \infty$?

(e) (5 points) Show that X_n has a stationary distribution if and only if $\mu > \lambda$.

(f) (5 points) Now we consider two tandem service stations. In detail, we assume the arrivals follow the same distribution as in the single-station case. Upon arrival, customers will first join station 1 with i.i.d. geometric service times with parameter μ_1. After finishing service in station 1, they will go to station 2 with i.i.d. geometric service times with parameter μ_2. Find out a sufficient condition for this system to have a stationary distribution without proof. (hint: by time reversibility, in long term, the departures from station 1 will have the same distribution as the arrivals.)

Solution:

(a) Let's denote by A_n and B_n the number of arrivals and departures at time n, respectively. Then A_n independently follows the Bernoulli distribution $Ber(\lambda)$. Since the service time of each customer follows a geometric distribution $Geo(\mu)$, the customer in service at each time will depart with probability μ independently (recall how to generate a geometric random variable). Hence, given $X_n \geq 1$, B_n also follows the Bernoulli distribution $Ber(\mu)$. But given $X_n = 0$, it's reasonable that $B_n = 0$. Hence, we have the following transition:

$$X_{n+1} = X_n - B_n + A_n.$$

Now verify the Markov property.

$$\begin{aligned}
&P(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= P(X_n - B_n + A_n = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= P(X_n - B_n + A_n = j \mid X_n = i) \\
&= P(X_{n+1} = j \mid X_n = i),
\end{aligned}$$

where the last second equation holds, since A_n doesn't rely on anything and B_n is only dependent on X_n.

Transition probabilities: For $X_n = 0$, we have

$$P_{0,j} = P(X_{n+1} = j \mid X_n = 0) = P(X_n - B_n + A_n = j \mid X_n = 0) = P(A_n = j \mid X_n = 0) = P(A_n = j),$$

and hence, $P_{0,0} = P(A_n = 0) = 1 - \lambda$, $P_{0,1} = P(A_n = 1) = \lambda$ and $P_{0,j} = 0$ for all $j \geq 2$.

For $X_n = i \geq 1$, we have

$$P_{i,j} = P(X_{n+1} = j \mid X_n = i) = P(A_n - B_n = j - i \mid X_n = i),$$

and hence, $P_{i,i-1} = P(A_n = 0)P(B_n = 1 \mid X_n = i) = (1-\lambda)\mu$, $P_{i,i+1} = P(A_n = 1)P(B_n = 0 \mid X_n = i) = \lambda(1-\mu)$, $P_{i,i} = 1 - P_{i,i-1} - P_{i,i+1} = \lambda\mu + (1-\lambda)(1-\mu)$ and $P_{i,j} = 0$ for all $j \leq i - 2$ or $j \geq i + 2$.

Note that this DTMC can only make transitions from a state to itself or one of its two nearest neighbors.

(b) In order to prove the time reversibility, one can use the proposition: if a Markov chain with transition probabilities P_{ij} and there exists $\pi_i \geq 0$ such that

$$\pi_i P_{i,j} = \pi_j P_{ji}, \quad \sum_i \pi_i = 1,$$

then the chain is time reversible and π are its stationary probabilities.

We can easily obtain the stationary probabilities for each state $i = 0, 1, 2, \ldots$ and $j = i + 1$. This yields

$$\begin{aligned}
\pi_0 \lambda &= \pi_1 (1-\lambda)\mu & \quad\Rightarrow\quad & \pi_1 = \frac{\lambda}{\mu(1-\lambda)}\pi_0 \\
\pi_1 \lambda(1-\mu) &= \pi_2 (1-\lambda)\mu & \quad\Rightarrow\quad & \pi_2 = \frac{\lambda(1-\mu)}{\mu(1-\lambda)}\pi_1 \\
&\vdots & & \vdots \\
\pi_i \lambda(1-\mu) &= \pi_{i+1} (1-\lambda)\mu & \quad\Rightarrow\quad & \pi_{i+1} = \frac{\lambda(1-\mu)}{\mu(1-\lambda)}\pi_i \\
&\vdots & & \vdots
\end{aligned}$$

Hence, in general, for any state $i \geq 1$,

$$\pi_i = \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \pi_1 = \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \frac{\lambda}{\mu(1-\lambda)}\pi_0.$$

Since $\sum_i \pi_i = 1$, we obtain

$$\pi_0 \left[1 + \sum_{i=1}^{\infty} \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \frac{\lambda}{\mu(1-\lambda)}\right] = 1,$$

or

$$\pi_0 = \left[1 + \sum_{i=1}^{\infty} \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \frac{\lambda}{\mu(1-\lambda)}\right]^{-1}. \tag{1}$$

In this setting, $\lambda = 0.6$ and $\mu = 0.9$,

$$\pi_0 = \left[1 + \sum_{i=1}^{\infty} \left(\frac{0.6 \times 0.1}{0.9 \times 0.4}\right)^{i-1} \frac{0.6}{0.9 \times 0.4}\right]^{-1} = \left[1 + \sum_{i=1}^{\infty} \left(\frac{1}{6}\right)^{i-1} \frac{5}{3}\right]^{-1} = \frac{1}{3}.$$

and for any state $i \geq 1$,

$$\pi_i = \left(\frac{0.6 \times 0.1}{0.9 \times 0.4}\right)^{i-1} \frac{0.6}{0.9 \times 0.4} \cdot \frac{1}{3} = \frac{5}{9 \times 6^{i-1}}.$$

From the proposition, the chain is time reversible and π are its stationary probabilities. Since π is a distribution, then the stationary distribution is π.

(c) First we need to use balance equations to get stationary probabilities:

$$\pi_j = \sum_i \pi_i P_{ij}, \quad \sum_i \pi_i = 1.$$

We can easily obtain the stationary probabilities for each state $j = 0, 1, 2, \ldots$. This yields

$$\begin{aligned}
&\pi_0 = \pi_0(1-\lambda) + \pi_1(1-\lambda)\mu && \Rightarrow \quad \pi_1 = \frac{\lambda}{\mu(1-\lambda)}\pi_0 \\
&\pi_1 = \pi_0\lambda + \pi_1(\lambda\mu + (1-\lambda)(1-\mu)) + \pi_2(1-\lambda)\mu && \Rightarrow \quad \pi_2 = \frac{\lambda(1-\mu)}{\mu(1-\lambda)}\pi_1 \\
&\quad\vdots && \quad\vdots \\
&\pi_i = \pi_{i-1}\lambda(1-\mu) + \pi_i(\lambda\mu + (1-\lambda)(1-\mu)) + \pi_{i+1}(1-\lambda)\mu && \Rightarrow \quad \pi_{i+1} = \frac{\lambda(1-\mu)}{\mu(1-\lambda)}\pi_i \\
&\quad\vdots && \quad\vdots
\end{aligned}$$

Note that we can also use cut method to get the same result.

Hence, in general, for any state $i \geq 1$,

$$\pi_i = \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \pi_1 = \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \frac{\lambda}{\mu(1-\lambda)}\pi_0.$$

Since $\sum_i \pi_i = 1$, we obtain

$$\pi_0 \left[1 + \sum_{i=1}^{\infty} \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \frac{\lambda}{\mu(1-\lambda)}\right] = 1,$$

or

$$\pi_0 = \left[1 + \sum_{i=1}^{\infty} \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1} \frac{\lambda}{\mu(1-\lambda)}\right]^{-1}. \tag{2}$$

From (2), if $\lambda = 0.9$ and $\mu = 0.6$, the stationary probability $\pi_i = 0$ for all $i \geq 0$. But $\pi = 0$ is not a distribution, so X_n does not have a stationary distribution.

(d) X_n will go to infinity with probability 1 as $n \to \infty$.

(e) If $\mu > \lambda$, then $1 - \mu < 1 - \lambda$ and hence $\frac{\lambda(1-\mu)}{\mu(1-\lambda)} < 1$. Thus, $\sum_{i=1}^{\infty} \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1}$ converges, and we can get stationary distribution from (2). Otherwise, if $\mu \leq \lambda$, then $\frac{\lambda(1-\mu)}{\mu(1-\lambda)} \geq 1$, and $\sum_{i=1}^{\infty} \left(\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right)^{i-1}$ will go to infinity. From (2), the stationary probability $\pi_i = 0$ for all $i \geq 0$. But that's not a distribution, so X_n does not have a stationary distribution.

(f) $\lambda < \mu_1$ and $\lambda < \mu_2$.

Note that you can treat the tandem service station as two separate service stations. If you don't consider the station 2, the sufficient and necessary condition for the station 1 is $\lambda < \mu_1$. Due to time reversibility, in long term, the departures from the station 1 will have the same distribution as the arrivals, i.e., Bernoulli distribution with parameter λ. Now in order to make the station 2 has the stationary distribution, we just need $\lambda < \mu_2$.

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